

Original Report

Application of human factors analysis and classification system model to event analysis in radiation oncology



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Received 18 March 2014; revised 21 May 2014; accepted 26 May 2014

Abstract

Purpose: To assess the applicability of the human factors analysis and classification system (HFACS) model for analyzing events in a radiation oncology environment by comparing the HFACS analysis results between novices and experts.

Methods and materials: Four novices (resident physicians, 2-4 postgraduate years) volunteered to participate and were asked to independently perform the HFACS analysis on the selected 30 events. The events were divided into the following 2 sets: (1) the description of events was given in detail (eg, the conditions under which the events occurred including information about root causes of error); (2) the description of events was given without detailed information. Each novice's categorization of events into 4 main and 12 sublevels were compared with expert categorization ("gold standard") and used for statistical analysis. Analysis of variance was performed to test the difference in mean agreement with the expert-defined gold standard between the novices across the 4 main levels; and across the 12 sublevels independently.

Results: There were no significant differences in mean agreement with the expert-defined gold standard among the 4 novices (novice versus novice; $P > .05$) across the 4 main and 12 sublevels, respectively. There was a significant difference in mean agreement with the expert-defined gold standard among the 4 main levels and 12 sublevels ($P < .05$) across the 4 novices. For the 4 main levels, there was a significant difference in agreement with the expert-defined gold standard for events with detailed information versus events without detailed information provided across the 4 novices. The additional information did not improve mean agreement on the 12 sublevels.

Conclusions: Novices learned to use the HFACS model for higher level analysis (4 main levels) with 1 hour training. Regardless of the amount of detail provided in the event description, the study results indicate a need of formal training for novices to better understand the definition and their interpretation at the 12 sublevel analyses.

Published by Elsevier Inc. on behalf of American Society for Radiation Oncology.

Supplementary material for this article (<http://dx.doi.org/10.1016/j.prro.2014.05.010>) can be found online at www.practicalradonc.org.

Conflicts of interest: None.

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<http://dx.doi.org/10.1016/j.prro.2014.05.010>

1879-8500/Published by Elsevier Inc. on behalf of American Society for Radiation Oncology.

Introduction

An event reporting system is a tool for capturing, analyzing, and learning about a broad range of potential safety issues.^{1,2} In aviation, systematic event reporting of crashes and near-misses is credited with helping to reduce the risk of fatal accidents by 73%.³ Similar improvements have been achieved in the nuclear power industry.⁴

The Patient Safety and Quality Improvement Act of 2005 enabled the creation of patient safety organizations (PSOs) to promote event reporting in health care as a means to improve patient safety. PSOs encourage health care organizations to voluntarily report and share quality and patient safety information without fear of legal discovery (Agency for Healthcare Research and Quality).⁵ PSOs serve as independent, external experts who can collect, analyze, and aggregate clinical data locally, regionally, and nationally to develop insights into the underlying causes of patient safety events.

The American Society for Radiation Oncology and the American Association of Physicist in Medicine have jointly partnered with Clarity (a pioneer in the PSO movement since 2002)⁶ to create a radiation oncology-specific PSO, called the radiation oncology-incident learning system (RO-ILS).⁷

Event reporting systems require systematic means to categorize events (eg, with regard to what, why, etc) to enable data pooling. There are many methods that have extensively been used for analyzing events in certain settings and have limited application in others (see Appendix A for summary; available online only at www.practicalradonc.org). The human factors analysis and classification system (HFACS) model is perhaps the most widely used method, having been applied in civil and military aviation,^{8,9} helicopter maintenance,¹⁰ coal mining,¹¹ rail transportation,¹² maritime,¹³ construction,¹⁴ health care,¹⁵ and road transportation.¹⁶

The HFACS was developed by Shappell and Wiegmann,¹⁷ based on Reason's "Swiss cheese model" (Fig 1) of human error causation of latent and active.¹⁸ HFACS investigates events using 4 main levels of failure, and 12 additional sublevels (Fig 2; see Appendix B for definitions; available online only at www.practicalradonc.org).

To categorize an event into the various categories of the HFACS model, a user starts at the first main level (unsafe act, mandatory to select), then proceeds to the corresponding sublevels (eg, unintended violation vs intended violation), followed by the nano-codes (unintended violation: skill based vs decision-based vs perceptual; intended violation: routine violation vs exceptional violation). The "actual" selection of a nano-code is an iterative decision-making process between the main level, sublevels, and nano-codes, which is typically dependent on the user's knowledge and experience in using the HFACS model. Next, the user follows the same method as

described above to categorize the remaining 3 main levels and their corresponding sublevels and nano-codes (not mandatory; select as appropriate). The categorization on the nano-codes and sublevels is essentially a more detailed (and more specific) assessment compared with the 4 main levels, and hence these are not independent of each other.

Pooling data from multiple centers raises the question of reliability, especially as many radiation oncology personnel do not have formal training in error analysis. Data from several studies,⁸⁻¹⁶ suggest that the HFACS system can be consistently applied by different users.

This, and evidence from previous studies, indicate that the HFACS model can be considered a reasonable and reliable event analysis method for standard interpretation of the data to be collected by the RO-ILS. Indeed, the formulation of the error reporting in the RO-ILS system is quite similar to the HFACS approach.

The primary goal of this study was to assess the reliability of the HFACS model for analyzing radiation oncology events between novice evaluators (using expert review as the gold standard). We also assessed the impact of the degree of detail provided to the evaluator on the reliability of the HFACS model.

Methods and materials

In June 2012, the department of radiation oncology at the University of North Carolina at Chapel Hill implemented a "Good Catch" event reporting program.¹⁹ Through January 2014, 450 events were reported and analyzed.

Experts (2 industrial engineers with formal training and experience using the HFACS model) selected 30 radiation oncology events from the department database (5 representative samples are provided in Appendix C; available online only at www.practicalradonc.org). Using an iterative decision-making process, these 2 experts jointly reviewed each event and categorized them (by consensus) into the 4 main and 12 sublevels of the HFACS model. Although the event categorizations on nano-code were also done, due to the small sample of events considered for analysis, the categorization on 12 sublevels and 4 main levels were considered for comparison and data analysis. For the expert's analysis, the 4 main level and 12 sublevel categorizations were verified to be consistent with each other. Any disagreement between the experts were discussed and resolved to a consensus. The experts' categorization was considered as a "gold standard."

Four novices (resident physicians, 2 to 4 postgraduate years) volunteered to participate and were asked to independently perform the HFACS analysis on the selected 30 events. Novices performed event categorization into 12 sublevels first. Experts then converted the 12 sublevels categorization into the 4 main level categories. The novices' data on events categorization were compared

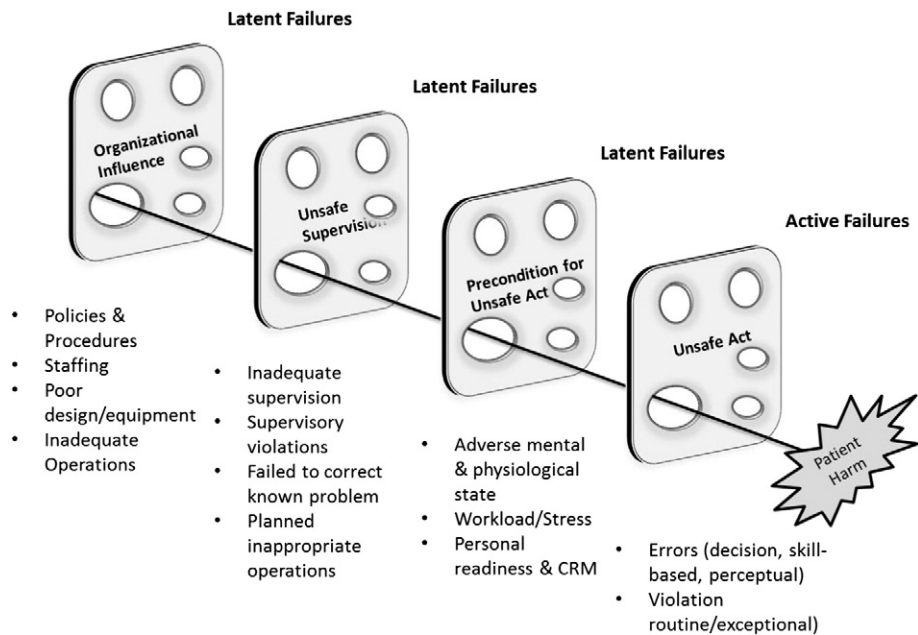


Figure 1 Swiss-cheese model. Adapted with permission from Reason J.¹⁸ (CRM, crew resource management.)

with the gold standard (coded as 1 for agreement; 0 for no agreement; and averaged across 30 events to calculate mean agreement). The mean agreement with the expert-defined gold standard was considered as the dependent variable for statistical analysis (Table 1). First, 1-factor analysis of variance (ANOVA) was performed to test the

difference in mean agreement with the expert-defined gold standard among the 4 novices across the 4 main levels (ANOVA #1, Table 1), and across the 12 sublevels (ANOVA #2, Table 1) of the HFACS model independently. Second, 1-factor ANOVA was also performed to test the difference in mean agreement with the expert-

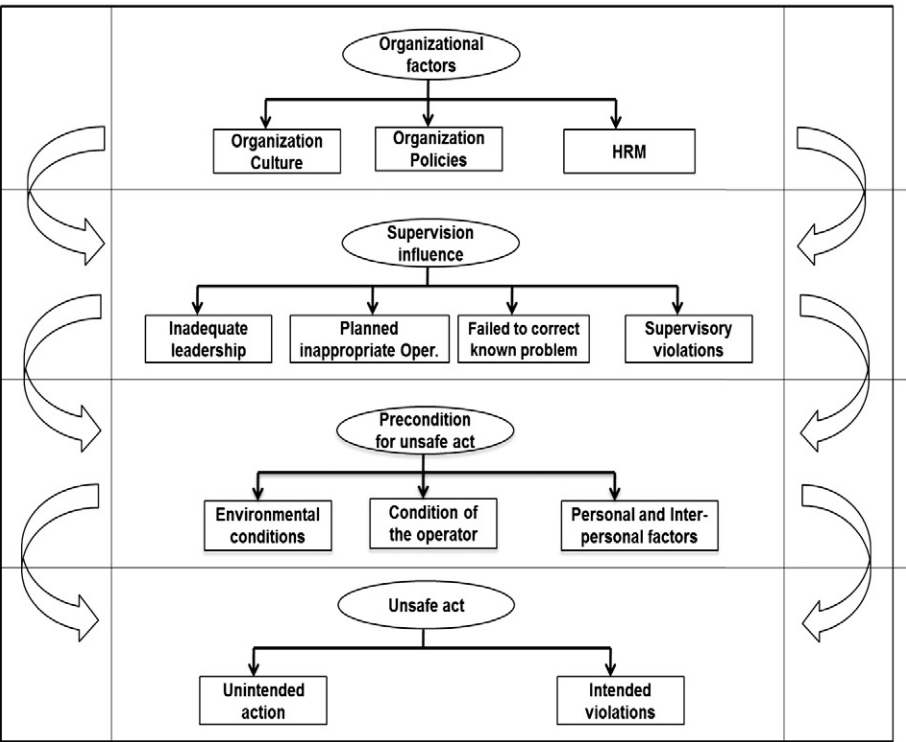


Figure 2 The human factors analysis and classification system model with categories, subcategories and hierarchy. (HRM, human research management.)

Table 1 Analysis of variance (ANOVA) test run in the study

ANOVA	Dependent variable (Mean agreement with the expert-defined gold standard)	Factor 1 (levels)	Factor 2 (levels)
#1	Averaged across 4 main levels	Novice (4)	
#2	Averaged across 12 sublevels	Novice (4)	
#3	Averaged across 4 novices	4 main levels (4)	
#4	Averaged across 4 novices	12 sublevels (12)	
#5	Averaged across 4 main levels	Novice (4)	Sets (2)
#6	Averaged across 12 sublevels	Novice (4)	Sets (2)
#7	Averaged across 4 novices	4 main levels (4)	Sets (2)
#9	Averaged across 4 novices	12 sublevels (12)	Sets (2)

defined gold standard between the 4 levels (ANOVA #3, Table 1), and 12-sublevels (ANOVA #4, Table 1) across 4 novices independently.

The events were divided into 2 sets. In the first set, the description of events was given in detail; for example, the conditions under which the events occurred including information about root causes of error. The second set consisted of a brief report of events without information on root causes. The analysis was repeated separately for these 2 sets, each of 15 events. For this set-wise comparison, first, a 2-factor ANOVA was performed to test the difference in mean agreement with the expert-defined

gold standard (averaged across 15 events in each set) between 4 novices and the 2 sets across the 4 main levels (ANOVA #5, Table 1) and 12 sublevels (ANOVA #6, Table 1), respectively (ie, to assess if the results varied with the amount of detail provided to the novices). Second, a respective 2-factor ANOVA was also performed to test the difference in mean agreement with the expert-defined gold standard between the 2 sets and the 4 main levels (ANOVA #7, Table 1); mean agreement between the 2 sets and the 12 sublevels (ANOVA #8, Table 1).

All significant ANOVA tests were followed by post hoc analysis using the Duncan multiple range test. An alpha value of 0.05 was used for significance for all the analyses.

Results

At the 4 main levels of the HFACS model, the average mean agreement with the expert-defined gold standard between 4 novices ranged from 66% to 97%. At the 12 sublevels, the average mean agreement with the expert-defined gold standard between 4 novices ranged from 50% to 82% (see Table 2).

The ANOVA #1 and ANOVA #2 results indicated no significant differences in mean agreement with the expert-defined gold standard between the 4 novices (novice vs novice) across 4 main levels ($P > .05$); and 12 sublevels ($P > .05$), respectively. There was a significant difference in mean agreement with the expert-defined gold standard among the 4 main levels (ANOVA #3; $P < .05$) and 12 sublevels (ANOVA #4; $P < .05$), respectively, across the 4 novices (Table 2). The post hoc analysis using the Duncan multiple range test indicated that, of the 4 main levels, there was significantly higher mean agreement with the

Table 2 Average mean agreement on main levels and sublevels of the HFACS model

4 main levels	No. ^a	Mean agreement ^b % (range)	F test ^c (<i>P</i> value)	12 sublevels	No. ^a	Mean agreement ^b % (range)	F test ^c (<i>P</i> value)
Organizational influence	13	72 (60-74)	F = 8.48 (<i>P</i> < .001)	Organizational culture	4	57 (50-58)	F = 61.1 (<i>P</i> < .001)
				Organizational policies	8	67 (50-71)	
				Human resource management	1	67 (50-71)	
Unsafe supervision	11	66 (61-71)		Inadequate supervision	4	58 (50-59)	
				Planned inappropriate operations	3	58 (50-75)	
				Failed to correct problem	3	70 (54-75)	
				Supervisory violations	1	50 (50)	
Precondition for unsafe act	20	84 (75-84)		Environmental factors	5	75 (60-83)	
				Condition of the operator	2	59 (56-62)	
				Personal Factors	13	80 (67-88)	
Unsafe act	29	97 (96-98)		Unintended Action	20	82 (81-83)	
				Intended violations	9	68 (65-71)	

^a Number of events with attribution to this level per the gold standard review.

^b Mean agreement with expert-defined gold standard.

^c F distribution test.

Table 3 Average mean agreement on main levels and sublevels for events with detail information versus those without detailed information

Main levels	Mean agreement on events % (range)			Sublevels	Mean agreement on events % (range)		
	With detailed information	Without detailed information	F test ^a (<i>P</i> value)		With detailed information	Without detailed information	F test ^a (<i>P</i> value)
Organizational influence	71 (56-78)	62 (56-72)	F = 17.7 (<i>P</i> < .001)	Organizational culture	57 (50-70)	50 (50)	F = 1.88 (<i>P</i> = .17)
Unsafe supervision	75 (64-93)	56 (50-60)		Organizational policies	55 (38-63)	53 (50-62)	
				Human resource management	49 (38-60)	38 (35-45)	
				Inadequate supervision	60 (50-70)	50 (50)	
Precondition for unsafe act	89 (75-95)	76 (75-76)	Planned inappropriate operations	59 (50-75)	57 (50-75)		
			Failed to correct problem	62 (50-75)	50 (50)		
			Supervisory violations	40 (25-47)	37 (35-50)		
			Environmental factors	63 (50-100)	73 (64-90)		
			Condition of the operator	59 (56-62)	53 (50-58)		
Unsafe act	99 (96-100)	96 (95-97)	Physical and mental limitations	69 (58-75)	69 (57-78)		
			Unintended action	69 (66-71)	86 (86)		
			Intended violations	58 (50-64)	74 (70-75)		

^a F distribution test.

expert-defined gold standard for “unsafe act” (97%) and “precondition for unsafe act” (82%) compared with “supervisory violations” (66%) and “organizational influence” (72%) across 4 novices ($P < .001$, Table 2). Of all the 12 sublevels, unintended violations of the unsafe act (82%), and personal factors of the precondition for unsafe act (80%) had significantly higher mean agreement with the expert-defined gold standard; all sublevels of supervisory violation and human resource management and organization culture of organization factors had significantly less mean agreement with the expert-defined gold standard compared with other sublevels ($P < .001$, Table 2) across 4 novices.

For set-wise analysis, the 2-factor ANOVA #5 and ANOVA #6 results indicated no significant differences in mean agreement with the expert-defined gold standard between the novices (novice vs novice) on the 2 sets (ie, events with detailed information vs events with no detailed information) across the 4 main levels ($P > .05$; Table 3); and 12 sublevels ($P > .05$; Table 3), respectively. For the 4 main levels ANOVA #7, there was a significantly greater level of agreement (ie, concordance) with the expert-defined gold standard for events with detailed information versus the events without detailed information ($P < .001$) across 4 novices. The post hoc analysis using the Duncan multiple range test indicated the mean agreement with the expert-defined gold standard was improved for precondition for unsafe act, supervisory violation, and organizational factors ($P < .001$) for events with detailed information versus events with no detailed information across 4 novices. This indicates that the detailed

information helped novices to correctly choose the higher main levels (root causes) of HFACS tool (Table 3). Conversely, For the 12 sublevels ANOVA #8, there was no significant difference in mean agreement with the expert-defined gold standard for events with detailed information versus the events without detailed information ($P < .001$) across 4 novices ($P > .05$, Table 3).

A post hoc analysis using the Duncan multiple range test indicated the mean agreement improved for precondition for unsafe act, supervisory violation, and organizational factors ($P < .001$) for events with detailed information versus events with no detailed information across 4 novices.

Discussion

The aim of the current study was to assess the applicability of the HFACS model to analyzing events in the radiation oncology environment. Two experts and 4 novices participated in an analysis of 30 reported events from the Department of Radiation Oncology at University of North Carolina Chapel Hill. The main findings of the current study were the following: (1) There was no difference in mean agreement with the expert-defined gold standard among the 4 novices on 4 main levels and 12 sublevels; (2) the agreement on the 4 main levels improved significantly for events with additional information versus events with no additional information across the 4 novices.

The findings from this study suggest that the HFACS model can reliably be used for the event analysis in the radiation oncology environment. With a minimal amount of training, the novice evaluators with no or minimal level of experience in human factors, were able to categorize events into the 4 main levels of the HFACS model. Also, increasing the level of detail in the events description improves the accuracy of HFACS categorization; eg, RO-ILS.

However, on the 12 sublevel categorization, there was a relatively suboptimal degree of agreement (ie, less concordance) for all events, with or without the detailed information. This suggests that the categorization task at the more specific 12 sublevel analysis should not be delegated to novices entering data into RO-ILS. More training is likely needed for the personnel performing the 12 sublevel analyses for accurate reporting to RO-ILS. Learning from other industries (eg, aviation, construction, etc) the percent agreement of interrater (between raters) reliability on any human error classification tool of 65% and above is considered acceptable and reliable.^{14,20-22} The current study shows an average mean agreement of 80% (range, 66%-97%) on 4 main levels, and average mean agreement of 65.9% (range, 50%-82%) on 12 sublevels. These findings suggest that the HFACS model can be considered as a reliable tool to be implemented in RO-ILS.

There are inherent limitations to this study. First, the results were based on a small sample of events with unequal representation of all sublevels of the HFACS model. For example, there was only 1 event categorized under "human resource management" and 2 events under "organization culture" of the "organizational influence" level. Nevertheless, this is the first study in our field to address this issue and the events considered represent commonly occurring incidents in radiation therapy (eg, pacemaker issues, resimulation due to poor communication, avoidable replans, etc). Second, there may have been some discussions between the novices that may have increased agreement between the novices and affected the results. However, clear instructions were provided to the novices to not discuss their categorizations with each other and we have no evidence that they did communicate with each other. Also, in the discussions between the experts and the novices occurring after data collection, the novice's explanations and reasons for choosing each level and sublevel (as subjectively assessed by experts) varied between novices. This suggests that there was not much discussion between the novices and that their categorizations were each independent selections.

Directions for future research include the following: (1) A larger incident sample with comparatively equal representation of all levels and the sublevels of the HFACS. This will enable one to better assess the reliability of the HFACS model at the nano-code level (see Appendix A [available as supplementary material at www.practicalradonc.org] for description of nano-codes); and (2) a larger participation of all the stakeholders from other

groups within radiation oncology. This will enable us to better assess which groups of workers within radiation oncology might be most reliable participants in the categorization of events reported to RO-ILS.

Despite the limitations, this study is a reasonable first step to provide some confidence that the HFACS can be reliably used for analysis of events in the radiation oncology environment.

Summary

Thirty radiation oncology department's events database were chosen for root-cause analysis using HFACS model. Novice raters were given 1 hour training. Novice ratings were compared with that of experts' defined "gold standard." We noted that the HFACS model can be used by the novice evaluators for the analysis of events from the radiation oncology environment with proper training.

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